Additional file 4: STROBE checklist for cross-sectional studies

Separating definition from denominator in NHANES diabetes prevalence estimates: a cross-sectional, design-based comparison of HbA1c and fasting glucose surveillance definitions

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Item numbers and topics follow the STROBE checklist for cross-sectional studies (von Elm et al., Lancet 2007). Locations name the section of the manuscript, or of Additional file 1, where each item is reported. This is a descriptive methodological study of prevalence estimators: it estimates no exposure-outcome association, so items about confounding and adjusted estimates are marked as not applicable.

Item	Topic	Location in the manuscript
1a	Study design in the title or abstract	Title ("cross-sectional, design-based comparison"); Abstract, Methods
1b	Informative and balanced abstract	Abstract
2	Scientific background and rationale	Background, paragraphs 1 to 4
3	Objectives	Background, last paragraph
4	Key elements of the study design	Methods, Data source; Analysis plan; Configurations and contrasts
5	Setting, locations, and dates of recruitment and data collection	Methods, Data source (August 2021 to August 2023 and 2017 to March 2020 releases; access date)
6a	Eligibility criteria, sources, and methods of selecting participants	Methods, Analysis domains and weights; Results, Participants and domains; Additional file 1, Table S3
7	Outcomes, exposures, and other variables; diagnostic criteria	Methods, Phenotypes (thresholds, codes, boundaries); phenotype specification in Additional file 2
8	Data sources and measurement for each variable	Methods, Data source; Phenotypes; Limitations (HbA1c instruments)
9	Efforts to address potential sources of bias	Methods, Analysis domains and weights (documented weights for each domain); Sensitivity analyses; Verification (external reconciliation); Limitations
10	How the study size was arrived at	Methods, Data source (all eligible participants in the public files); Results, Participants and domains; Additional file 1, Table S3
11	Handling of quantitative variables	Methods, Phenotypes (thresholds with inclusive boundaries, published precision); Analyses not prespecified (cut points 6.4% and 6.6%)
12a	Statistical methods	Methods, Configurations and contrasts; Estimation
12b	Subgroups and interactions	Methods, Estimation (paired differences by age, sex, and race and Hispanic origin); Additional file 1, Table S6 and Figure S1
12c	Missing data	Methods, Analysis domains and weights; Estimation (no imputation, no ad hoc reweighting); Results, Participants and domains; Additional file 1, Table S4
12d	Analytic methods for the sampling strategy	Methods, Estimation (strata, PSUs, weights, Taylor linearization, Korn-Graubard intervals); Configurations and contrasts (joint linearization)
12e	Sensitivity analyses	Methods, Sensitivity analyses; Analyses not prespecified
13a	Numbers of individuals at each stage	Results, Participants and domains; Additional file 1, Table S3 (both releases)
13b	Reasons for non-participation at each stage	Additional file 1, Tables S3 and S4
13c	Flow diagram	Additional file 1, Table S3 (participant flow given as a table for both releases)
14a	Characteristics of participants	Additional file 1, Tables S4 and S6 (unweighted n by age, sex, and race and Hispanic origin)
14b	Participants with missing data for each variable	Results, Participants and domains; Figure 1; Additional file 1, Tables S3 and S4

Item	Topic	Location in the manuscript
15	Numbers of outcome events or summary measures	Table 1 (positive n and weighted percent for each configuration); Table 3
16a	Unadjusted estimates with precision; confounders	Tables 1 to 3 (weighted percentages and differences with 95% intervals); no adjustment for confounders (not applicable to prevalence estimation)
16b	Category boundaries for continuous variables	Methods, Phenotypes (HbA1c at or above 6.5%, fasting glucose at or above 126 mg/dL)
16c	Translation of relative risk into absolute risk	Not applicable (no relative measures are reported)
17	Other analyses (subgroups, interactions, sensitivity)	Results, Sensitivity analyses; Groups; Replication in 2017 to March 2020; The same adults, two thresholds (analyses not prespecified); Additional file 1, Tables S6 and S7
18	Key results with reference to the study objectives	Discussion, paragraph 1
19	Limitations, including sources of potential bias or imprecision	Discussion, Limitations
20	Cautious overall interpretation	Discussion, paragraphs 2 to 4; Conclusions
21	Generalisability	Discussion, paragraph 2 (populations and groups); Limitations
22	Funding and the role of funders	Declarations, Funding